

Electronic and Vibrational Spectra of Trans-1,3-Butadiene

Matthew Hale

East Carolina University

September 17, 2025

INTRODUCTION

Quantum mechanics provides chemists with a theoretical framework to predict the behavior of molecules at the atomic level. Computational chemistry programs, such as Gaussian 09, apply these principles without requiring the manual solution of Schrödinger's equation. GaussView, a graphical user interface (GUI) for Gaussian, allows for the construction of molecules, visualization of orbitals, and analysis of vibrational and electronic properties.

The molecule studied in this experiment was trans-1,3-butadiene, serving as a model system to understand π -electron delocalization. Using GaussView, this molecule is useful to visualize electron distribution and how it influences molecular orbitals and absorption spectra. The goal of this experiment was to become more acquainted with GaussView and Gaussian09 to examine molecules. Specifically, the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), then analyzing vibrational modes and infrared spectra, and finally computing the electron absorption spectrum. These values were then compared to the literature values to determine the accuracy of the method.

METHODS

This experiment was performed using Gaussian 09 on a Linux distribution (Red Hat), with GaussView as the molecular visualizer. A new directory was created using the mkdir command, and navigation through this directory was carried out with the cd command. A scheme file (pbe1_scheme.txt), which specifies the density functional and base set, was copied into the directory to standardize the calculations.

Trans-1,3-butadiene was constructed in GaussView by placing four carbons in a line, then the required hydrogen atoms and double bonds. The structure was saved as a Gaussian input file (.gjf). Geometry optimization was performed using density functional theory (DFT) with the PBE1 and 6-31+G(d,p) set. This minimizes the total electronic energy to obtain the most stable conformation of the molecule. A frequency calculation was run on the optimized geometry, confirming that the structure was the lowest energy; this also generated a vibrational spectrum. The output produced .log and .chk files. The .chk files contain the optimized wavefunction.

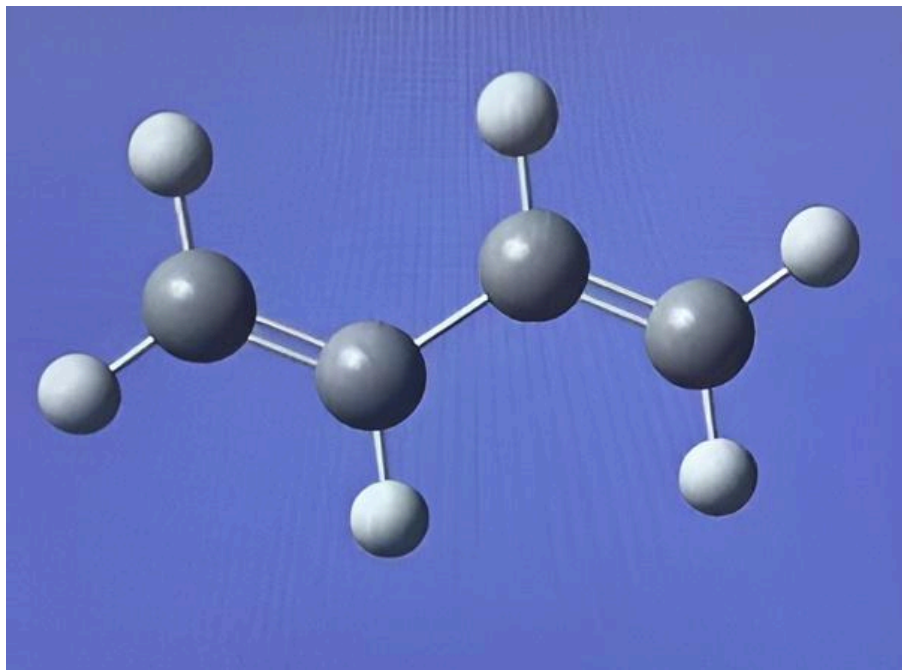
Molecular orbitals were visualized by loading the checkpoint files, then rendering the highest molecular orbital (HOMO) and the lowest molecular orbital (LUMO). Natural localized molecular orbitals (NLMOs) were then generated to compare bonding descriptions with the canonical orbitals.

Lastly, the electronic absorption spectrum was simulated using the time-dependent density functional theory (TD-DFT). The absorption maximum (λ_{max}) was calculated and compared to the literature values for the molecule.

RESULTS

Trans-1,3-butadiene was optimized, and this geometry is shown in **Figure 1**. The structure is planar due to the conjugation of the π -system across the carbon atoms.

Figure 1. Optimized geometry of trans-1,3-butadiene.



The molecular orbitals were viewed in GaussView. **Figure 2** shows the HOMO, which is spread out over the conjugated π -system. **Figure 3** shows the LUMO, which has antibonding characteristics and a nodal plane running between the carbons.

Figure 2. Homo of trans-1,3-butadiene.

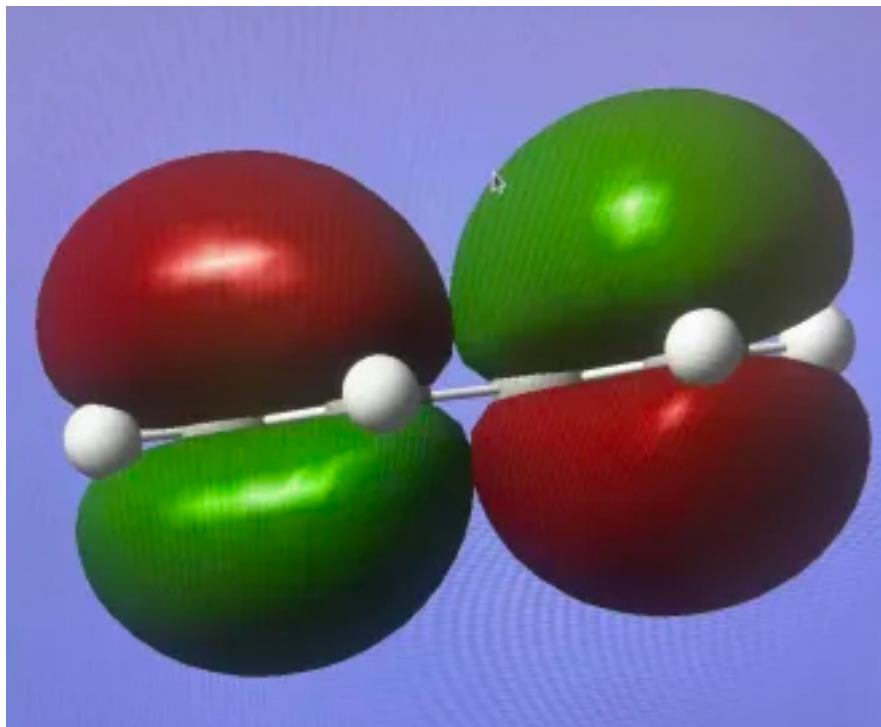
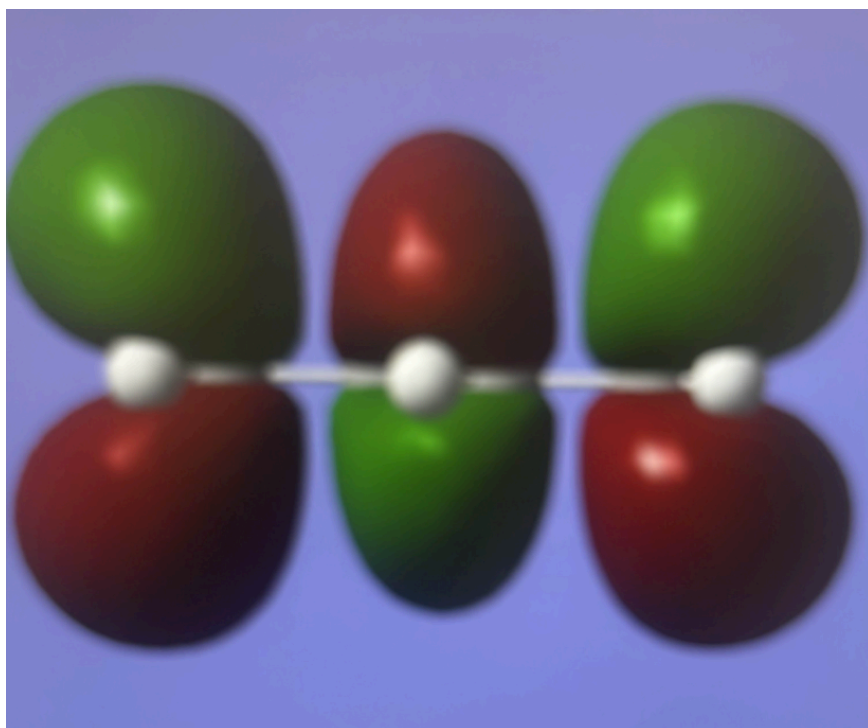
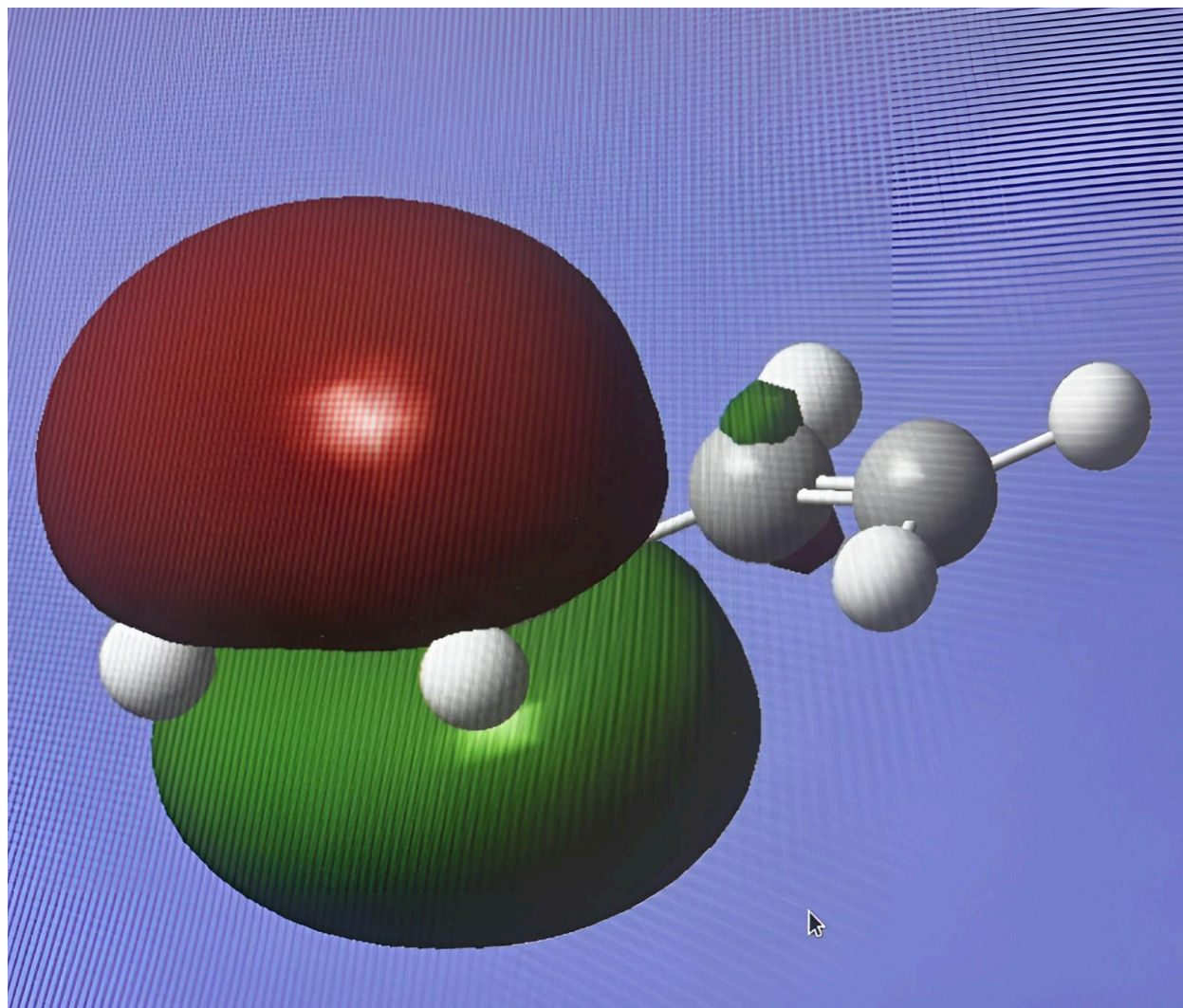


Figure 3. LUMO of trans-1,3-butadiene.



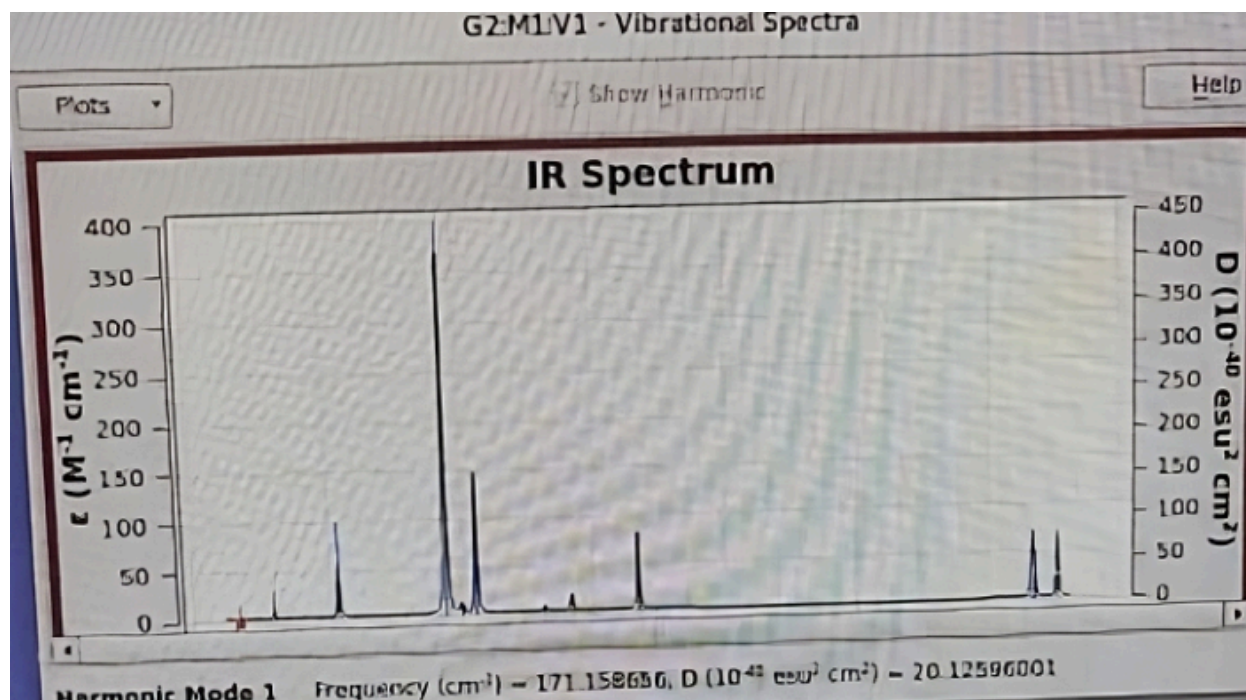
NLMOs were generated for comparison. **Figure 4** shows that these orbitals are localized to specific bonds. Although there are two NLMOs, they look very similar. This is because they are centered on double bonds, and trans-1,3-butadiene is symmetrical.

Figure 4. NLMOs of trans-1,3-butadiene.



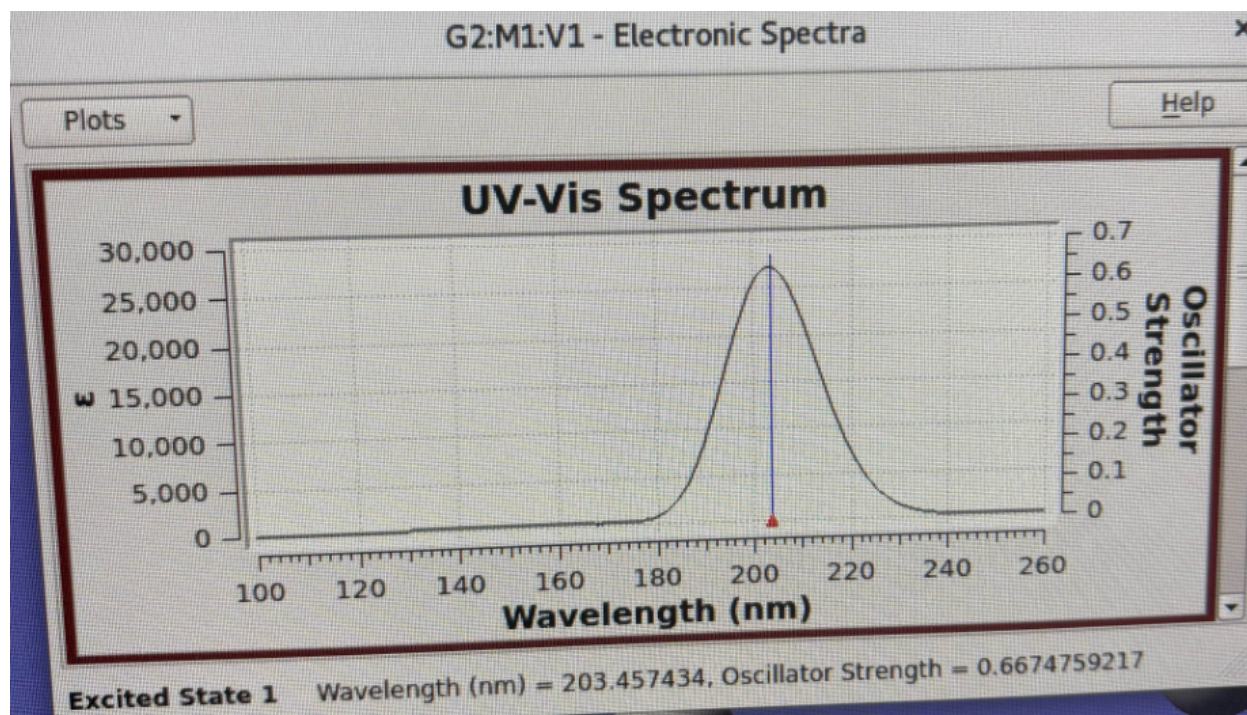
A vibrational frequency calculation was used to confirm that the optimized geometry was the actual lowest energy conformation, and also resulted in a vibrational spectrum (**Figure 5**).

Figure 5. Vibrational spectrum of trans-1,3-butadiene.



The electronic absorption spectrum was simulated (**Figure 6**). The calculated absorption maximum (λ_{max}) was found to be 203nm with an oscillator strength of 0.667.

Figure 6. Electronic spectra of trans-1,3-butadiene.



Discussion:

The optimized structure of trans-1,3-butadiene is planar, which makes sense because the π bonds are conjugated across the molecule. The HOMO spread across the entirety of the π -system whereas the LUMO showed antibonding on the nodal plane. This is exactly what is predicted for a conjugated diene.

While viewing the NLMOs, it is apparent they display partially the same bonding as the HOMO. They localize on the double bonds which mirror half of the HOMO with the other NLMO focused on the other double bond. These are just two different ways to represent the same electron density, in which one is delocalized and one is bond specific.

The vibrational spectrum gave the expected characteristics of a C=C double bond, as well as C-H stretching. Animating the molecule in GaussView also helped showcase these bonds and visualize what is associated with these peaks.

The electronic absorption spectrum gave a λ_{max} of 203 nm. The literature value for trans-1,3-butadiene is reported at 217 nm (Libretext, 2025). The value calculated by Gaussian 09 is close, but a difference can be found with the basis and functional set used for the calculations. Experimental effects are also not taken to account in the simulation. This value still showcases computational chemistry's efficacy when compared to experimental data.

CONCLUSION:

Computational chemistry provides a quick, cost-efficient method to estimate a rough range of experimental values and visualize molecular properties. While it is not perfect, and there are small differences that arise depending on the base and functional set, or experimental conditions present in an actual experiment, the results are still a good estimate of what can be expected in the lab. By using GaussView to analyze trans-1,3-butadiene, the ability to visualize the molecule became convenient, which makes programs like Gaussian 09 a powerful tool.

GLOSSARY

Basis Set - A collection of mathematical functions to describe orbitals in quantum calculations.

Canonical orbitals - Delocalized molecular orbitals obtained from quantum chemistry calculations (ex. HOMO and LUMO)

CD command - A linux command standing for change directory. Useful for navigating into different folders in the command line.

Checkpoint file (.chk) - A file type generated by Gaussian that stores the wavefunction as well as the optimized geometry of the molecule.

Density Functional Theory (DFT) - Computational Quantum method that approximates electron correlation that doesn't use wavefunctions, but uses electron density.

Electron Absorption Spectrum - A plot showing how a molecule absorbs light at different wavelengths.

GaussView - A graphical user interface (GUI) used to build molecules for Gaussian 09.

Gaussian 09 - A computational chemistry software for better analyzing quantum interactions.

Highest Occupied Molecular Orbital (HOMO) - The molecular orbital containing the highest-energy electrons in a given molecule.

Linux - An open-source operating system offering more customizability than alternatives like MacOS or Windows.

Lowest Unoccupied Molecular Orbital (LUMO) - The lowest-energy molecular orbital that does not contain electrons.

Mkdir command - A linux terminal command used to make a new directory (make a new folder).

Natural localized Molecular Orbitals (NLMOs) - Orbitals localized around specific bonds or atoms, rather than around canonical orbitals.

Scheme File - A file with a specified functional and base set for Gaussian calculations.

Time-Dependent Density Functional Theory (TD-DFT) - An extension of DFT to study electronically excited states of absorption spectra.

Wavefunction - A mathematical way that describes the behavior of a molecule's electrons.

Linux Shortcuts:

Cd - Change directory, used to navigate folders.

Cp - Copy command. Used to copy files to another location.

Ls - List command that displays the contents of a directory.

Tab Key - auto-completes when typing a command or file name

Up Arrow (↑) - Duplicates previous command entered into the terminal.

Down Arrow (↓) - Moves forward with command history after using the up arrow.

Pwd - Print working display, shows full path of a directory.

Cat - Displays file contents.

Tail - Shows the last few lines of a file.

References

East Carolina University, Department of Chemistry. (2022). *Physical Chemistry II Laboratory CHEM 3961* (Version 5.0) [Laboratory manual].

Frisch, M. J., Trucks, G. W., & et, al. (n.d.). *G09*. Gaussian. <https://gaussian.com/glossary/g09/>

Gaussian 09W tutorial an introduction to computational chemistry using G09W and. (n.d.).

<https://barrett-group.mcgill.ca/tutorials/Gaussian%20tutorial.pdf>

Libretexts. (2020, August 11). *17.5: Electronic spectra- ultraviolet and visible spectroscopy*.

Chemistry LibreTexts.

https://chem.libretexts.org/Courses/Winona_State_University/Klein_and_Straumanis_Guided/17%3A_Conjugated_Pi_Systems_and_Pericyclic_Reactions/17.05%3A_Electronic_Spectra-_Ultraviolet_and_Visible_Spectroscopy

Libretexts. (2025, March 26). *7.2: Wave Functions*. Physics LibreTexts.

[https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_\(OpenStax\)/University_Physics_III_-_Optics_and_Modern_Physics_\(OpenStax\)/07%3A_Quantum_Mechanics/7.02%3A_Wavefunctions](https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/University_Physics_III_-_Optics_and_Modern_Physics_(OpenStax)/07%3A_Quantum_Mechanics/7.02%3A_Wavefunctions)

Microsoft. (n.d.). *Keyboard shortcuts for Linux* [PDF]. Visual Studio Code.

<https://code.visualstudio.com/shortcuts/keyboard-shortcuts-linux.pdf>